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010	XSPI MEM & uSDHC CONN
011	S32K5XX_B2BConn_1
012	S32K5XX_B2BConn_2
013	CURRENT MEAS & ACQ CONN

Revisions

Rev	Description	Date
X3	Internal release	5th Nov, 2024
A	Rev A Release	1st April, 2024

Automotive Product Group
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Austin, TX 78735-8598

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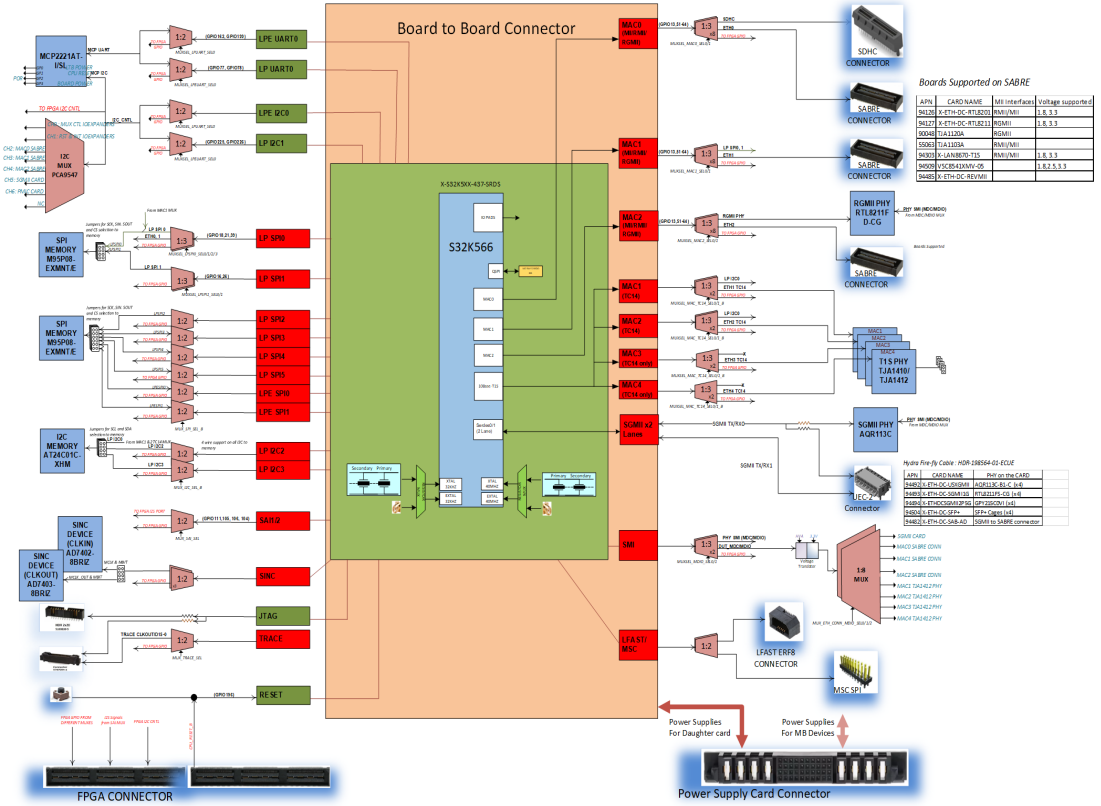
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Drawn by: <DrawnBy>	Page Title: 001-COVER PAGE	
Approved: <Approver>	Size C / Document Number SCH-94923 PDF: SPF-94923	Rev A
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DNP_&_JUMPER_Table

REF DES	JUMPER(DEFAULT)	PAGE NAME
J2	CLOSE 2-3	008 S32K5 uC PWR
J42	CLOSE 3-5, 4-6	008 S32K5 uC PWR
J44	CLOSE 2-3	010 XSPI MEM & uSDHC CONN

REF DES	ASSY_OPT	PAGE NAME
R175,R179,R186	DNP	004-S32K5 uC GPIO 1-127
R182,R184	DNP	006-S32K5 uC GPIO256-363
R138,R140,R141,R142,R171,Y1	DNP	007 S32K5 uC XTAL & SGMII
C3	DNP	008 S32K5 uC PWR
R147,R153,R154,R161	DNP	010 XSPI MEM & uSDHC CONN
R162,R163,R164	DNP	012 S32K5XX_B2BConn_2
R71,R75,R79,R83,R87,R91	DNP	013 CURRENT MEAS & ACQ CONN



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Drawing Title: **X-S32K5XX-437-DC**

Page Title: **002-BLOCK DIAGRAM**

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Stackup

Stackup Information:

Layer	Er	Info		Thickness
TOP		==========		0.5 oz+Plating=1.5mil
	3.2	PP	R5670G	2.7(mil)
L2		==========		1 Oz=1.2mil
	4.0	Core	370HR	3.15(mil)
L3		==========		0.5 Oz=0.6mil
	4.0	PP	370HR	7.87(mil)
L4		==========		1 Oz=1.2mil
	4.0	Core	370HR	4.0(mil)
L5		==========		0.5 Oz=0.6mil
	4.0	PP	370HR	6.5(mil)
L6		==========		1 Oz=1.2mil
	4.0	Core	370HR	3.15(mil)
L7		==========		1 Oz=1.2mil
	4.0	PP	370HR	8.2(mil)
L8		==========		1 Oz=1.2mil
	4.0	Core	370HR	3.15(mil)
L9		==========		1 Oz=1.2mil
	4.0	PP	370HR	6.5(mil)
L10		==========		0.5 Oz=0.6mil
	4.0	Core	370HR	4.0(mil)
L11		==========		1 Oz=1.2mil
	4.0	PP	370HR	7.87(mil)
L12		==========		0.5 Oz=0.6mil
	4.0	Core	370HR	3.15(mil)
L13		==========		1 Oz=1.2mil
	3.2	PP	R5670G	2.7(mil)
BOT		==========		0.5 oz+Plating=1.5mil

Finished:	77.94(7.8/-7.8) mil	1.98(+0.2/-0.2) MM
Designed:	78.75 mil	2.0 MM
Material:	MEG6+370HR	MEG6+370HR

Impedance Information:

Ctrl	Ref	Imp_type	Cust_req	Imp_req	HB_des	Imp_des	mask	H1	Er1	H2	Er2
L3	L2/L4	Differential	3.5/8.0	100+/-10%	3.5/8.0	99.77		3.15	4.0	8.47	4.0
L3	L2/L4	Differential	4.2/7.0	90+/-10%	4.2/7.0	90.57		3.15	4.0	8.47	4.0
L3	L2/L4	Single-Ended	3.8	50+/-10%	3.8	49.90		3.15	4.0	8.47	4.0
L5	L4/L6	Differential	4.8/7.0	90+/-10%	4.8/7.0	90.12		4.0	4.0	7.1	4.0
L5	L4/L6	Differential	4.0/8.5	100+/-10%	4.0/8.5	100.05		4.0	4.0	7.1	4.0
L5	L4/L6	Single-Ended	4.3	50+/-10%	4.3	50.08		4.0	4.0	7.1	4.0
L10	L9/L11	Differential	4.8/7.0	90+/-10%	4.8/7.0	90.12		4.0	4.0	7.1	4.0
L10	L9/L11	Differential	4.0/8.5	100+/-10%	4.0/8.5	100.05		4.0	4.0	7.1	4.0
L10	L9/L11	Single-Ended	4.3	50+/-10%	4.3	50.08		4.0	4.0	7.1	4.0
L12	L11/L13	Differential	3.5/8.0	100+/-10%	3.5/8.0	99.77		3.15	4.0	8.47	4.0
L12	L11/L13	Differential	4.2/7.0	90+/-10%	4.2/7.0	90.57		3.15	4.0	8.47	4.0
L12	L11/L13	Single-Ended	3.8	50+/-10%	3.8	49.90		3.15	4.0	8.47	4.0
L1	L2	Single-Ended	4.9	50+/-10%	4.9	50.03	Yes	2.7	3.2		
L1	L2	Differential	4.2/8.0	100+/-10%	4.2/8.0	100.12	Yes	2.7	3.2		
L1	L2	Differential	5.2/7.5	90+/-10%	5.2/7.5	90.12	Yes	2.7	3.2		
L14	L13	Single-Ended	4.9	50+/-10%	4.9	50.03	Yes	2.7	3.2		
L14	L13	Differential	4.2/8.0	100+/-10%	4.2/8.0	100.12	Yes	2.7	3.2		
L14	L13	Differential	5.2/7.5	90+/-10%	5.2/7.5	90.12	Yes	2.7	3.2		

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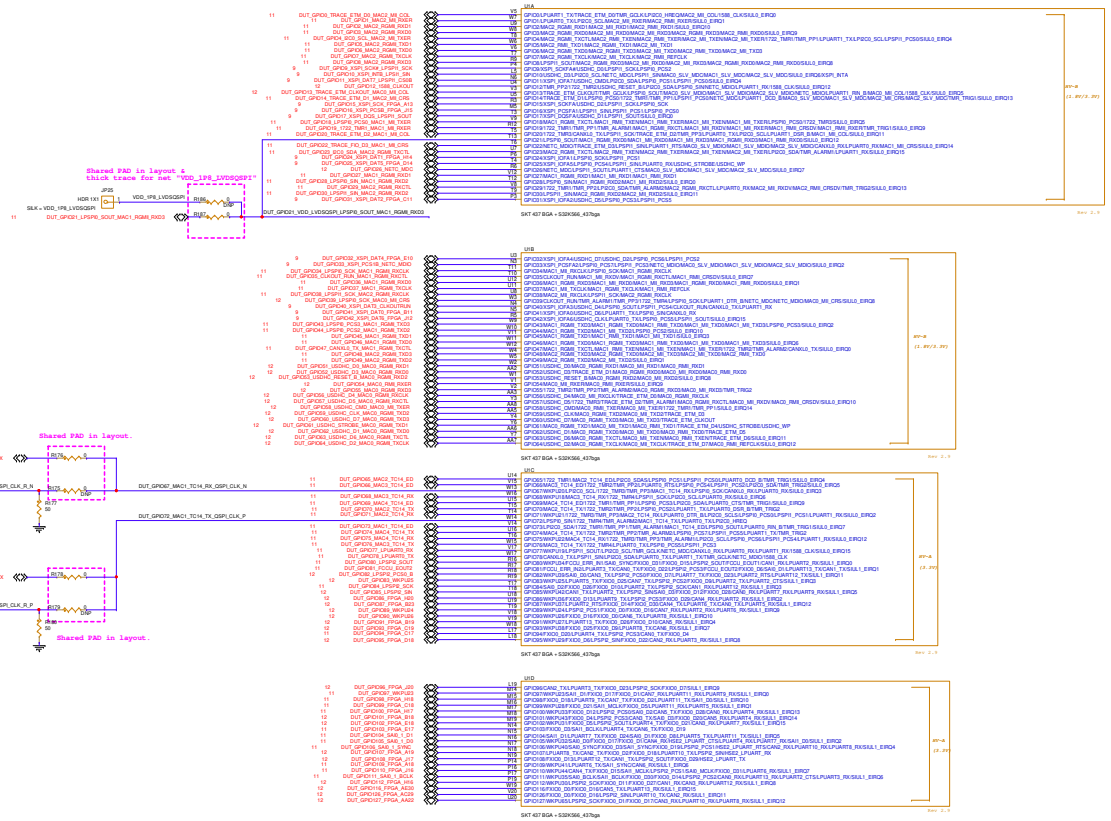
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Page Title: 003-STACKUP

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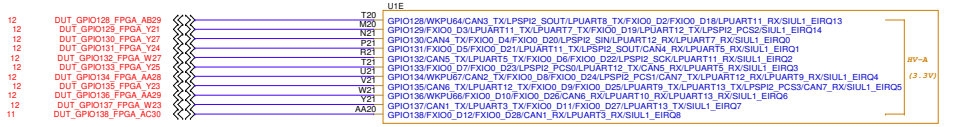
Date: Tuesday, April 01, 2025 | Sheet 3 of 13

S32K5 CONTOROLLER GPIO 1-127



Changes Since Conexus Internal Discussion:			
Drawing Title:	X-S32K5XX-437-DC		
Page Title:	004-S32K5 uC GPIO 1-127		
Size: D	Document Number: SCH-84923 PDF: SPF-84923		Rev: A
Date: Tuesday, April 01, 2025	Sheet 4 of 13		

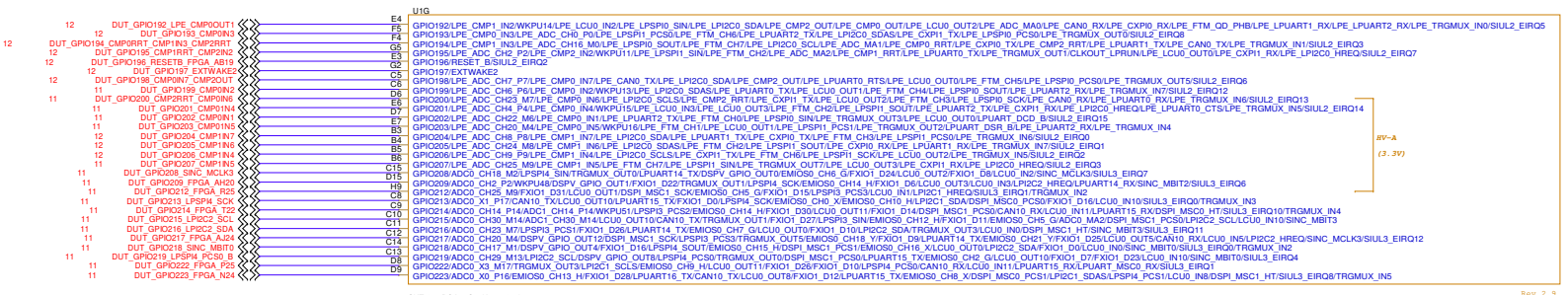
S32K5 CONTOROLLER GPIO 128-254



SKT 437 BGA + S32K566_437bga



SKT 437 BGA + S32K566_437bga



SKT 437 BGA + S32K566_437bga



SKT 437 BGA + S32K566_437bga

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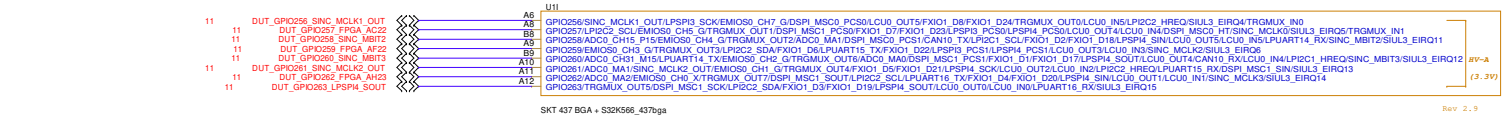
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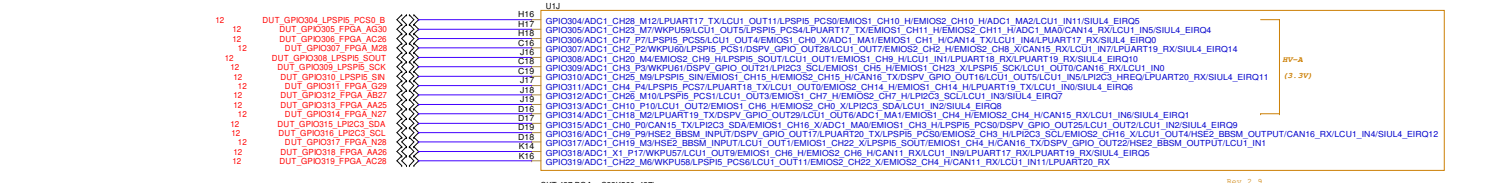
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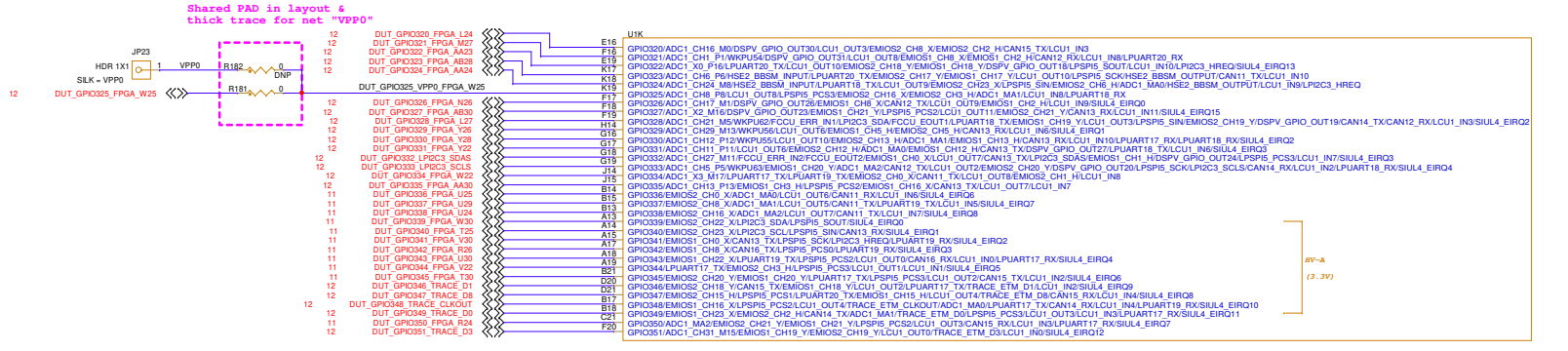
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S32K5 CONTOROLLER GPIO 256-363









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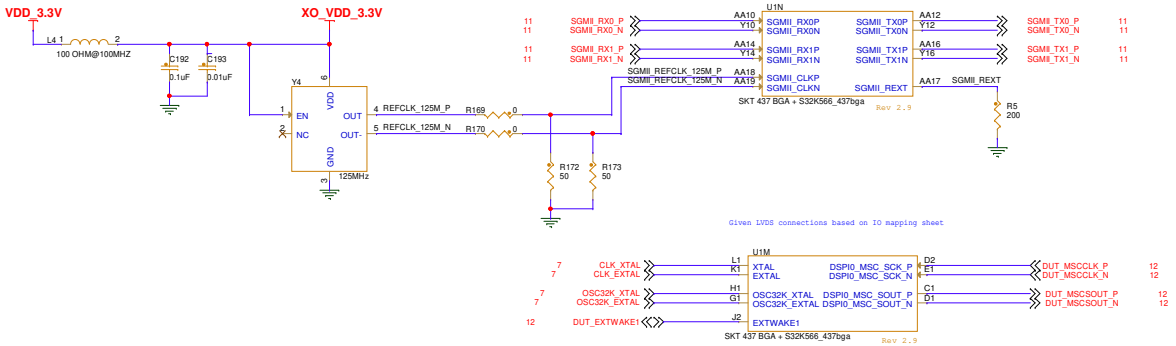
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Page Title: 006-S32K5 uC GPIO256-363

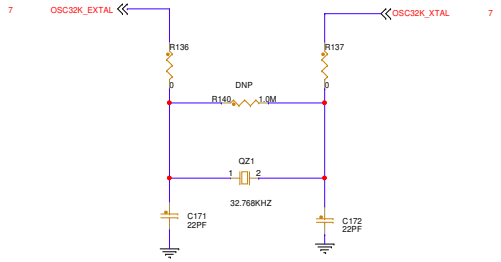
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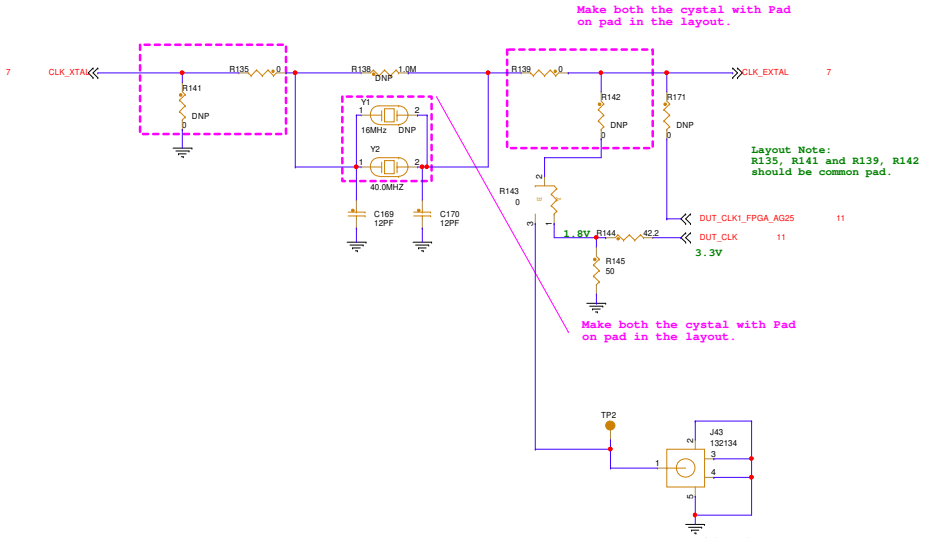
S32K5XX_IO#4



32 KHz

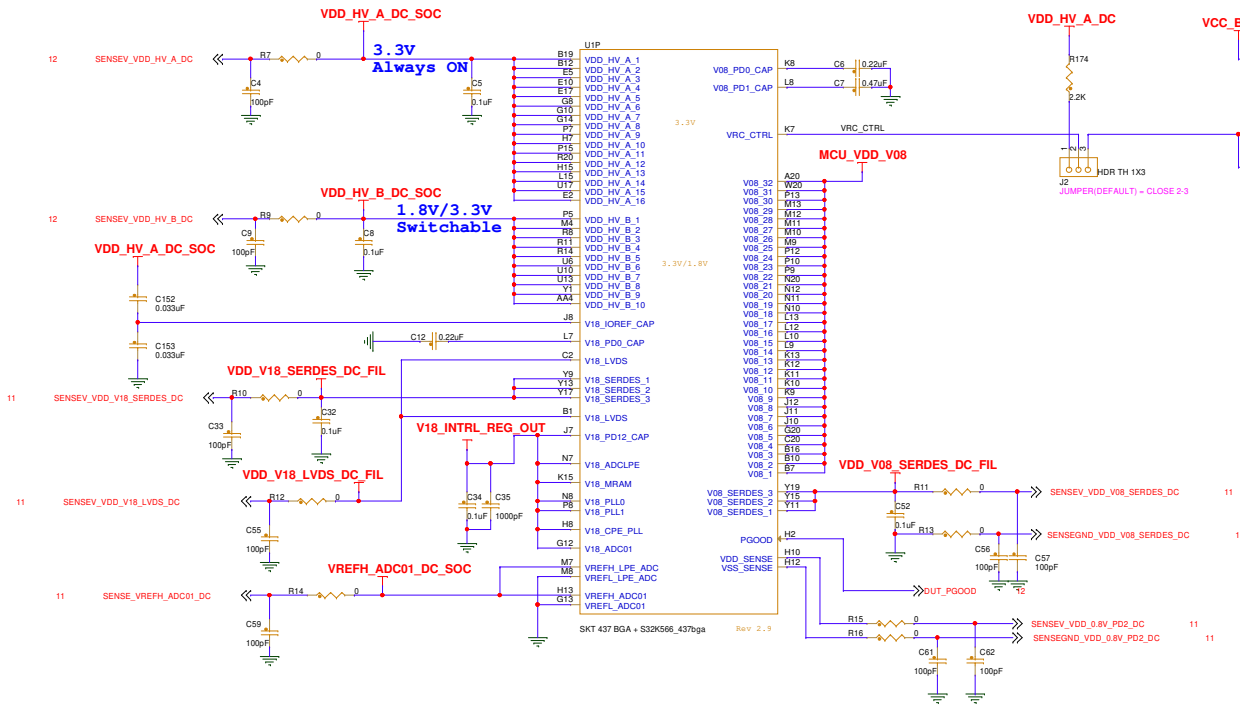


16MHz/40MHz Crystal

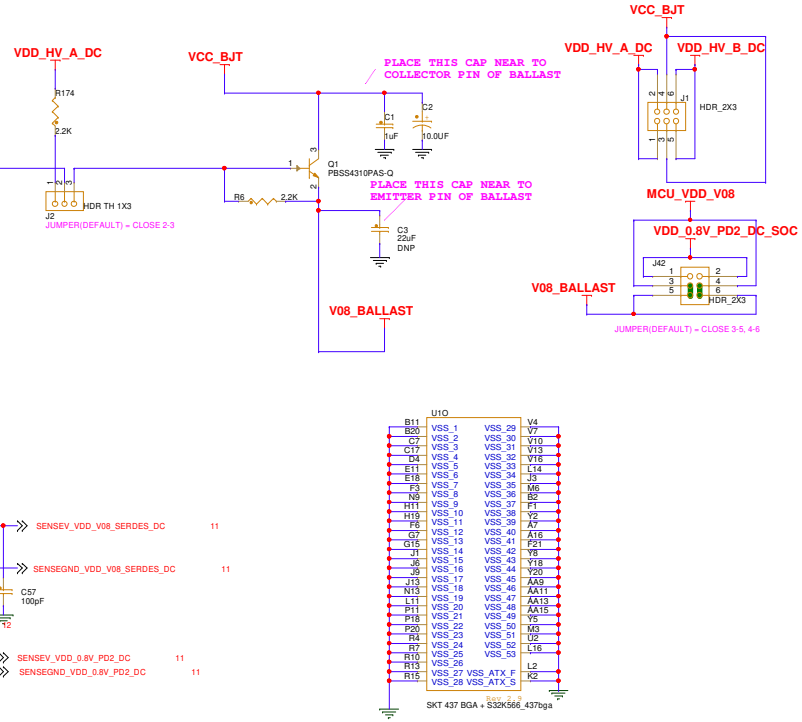


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Page Title: 007 S32K5 uC XTAL & SGMII		
Size C	Document Number SCH-94923 PDF: SPF-94923	Rev A
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MCU Power



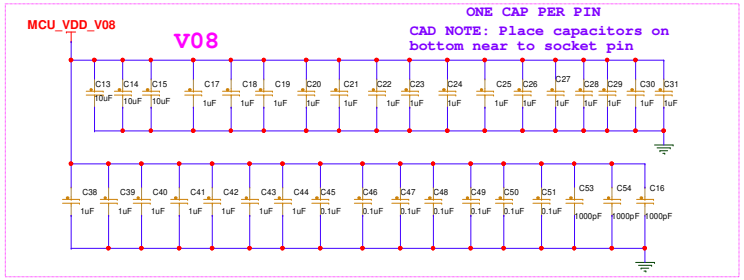
BALLAST CIRCUIT



V08

ONE CAP PER PIN

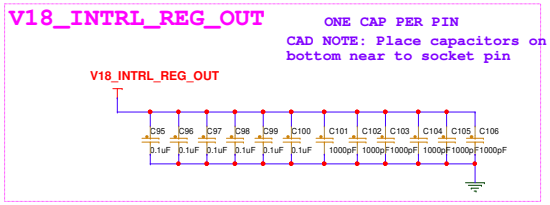
CAD NOTE: Place capacitors on bottom near to socket pin



V18_INTRL_REG_OUT

ONE CAP PER PIN

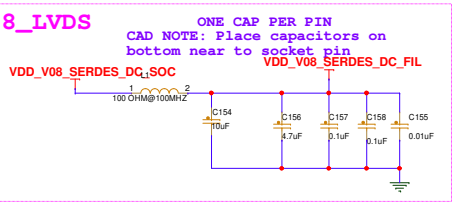
CAD NOTE: Place capacitors on bottom near to socket pin



V08_LVDS

ONE CAP PER PIN

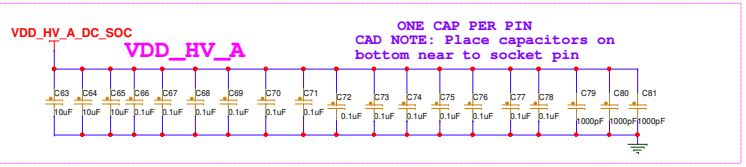
CAD NOTE: Place capacitors on bottom near to socket pin



VDD_HV_A

ONE CAP PER PIN

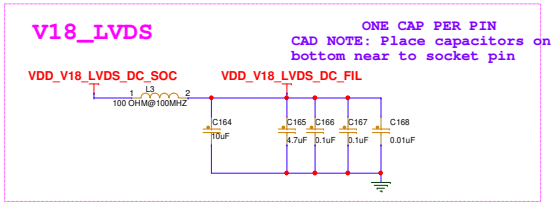
CAD NOTE: Place capacitors on bottom near to socket pin



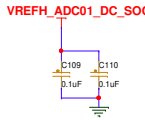
V18_LVDS

ONE CAP PER PIN

CAD NOTE: Place capacitors on bottom near to socket pin



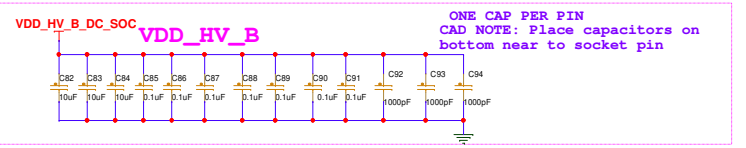
VREF_ADC



VDD_HV_B

ONE CAP PER PIN

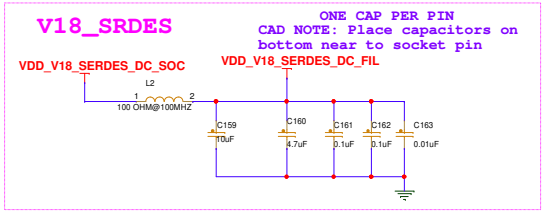
CAD NOTE: Place capacitors on bottom near to socket pin



V18_SRDES

ONE CAP PER PIN

CAD NOTE: Place capacitors on bottom near to socket pin



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Drawing Title: X-S32K5XX-437-DC

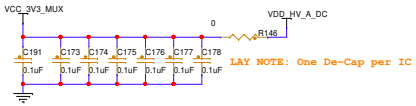
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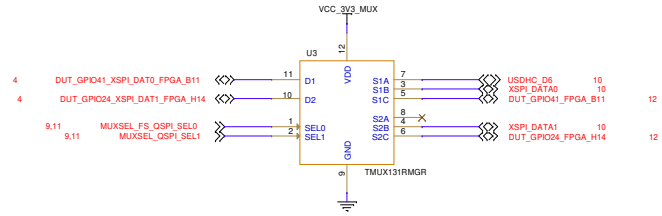
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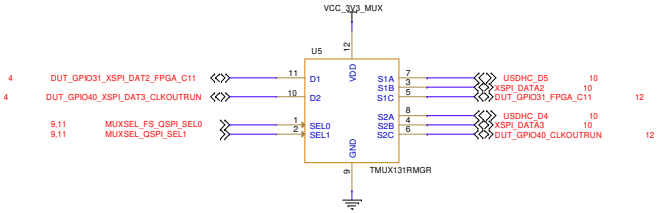
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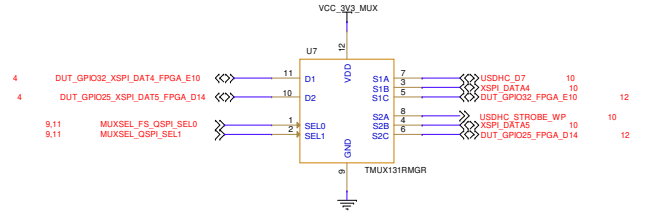
XSPI/SDHC MUX

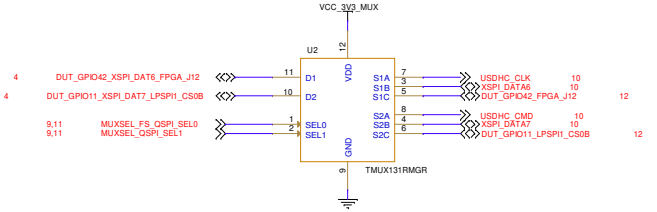
SEL1	SEL0	Selection
Low	Low	CH B => XSPI (8 bits)
Low	High	CH C => FPGA
High	Low	CH A => SDHC (8 bits)

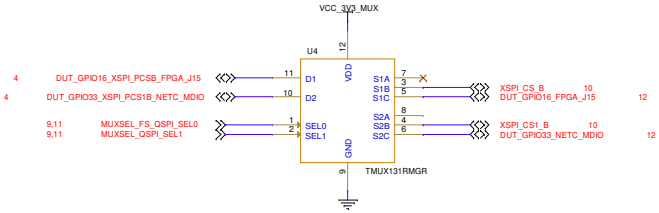


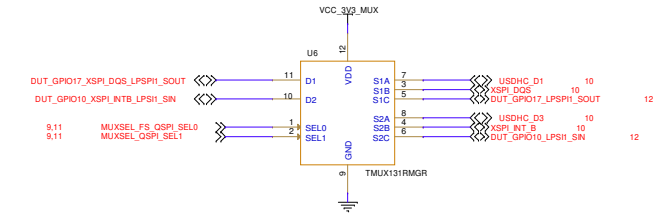


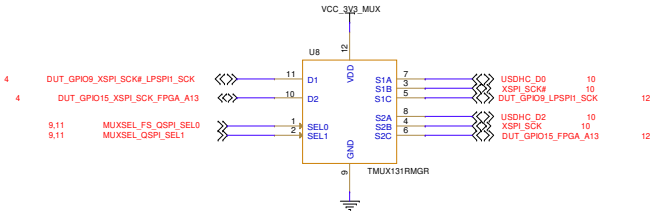












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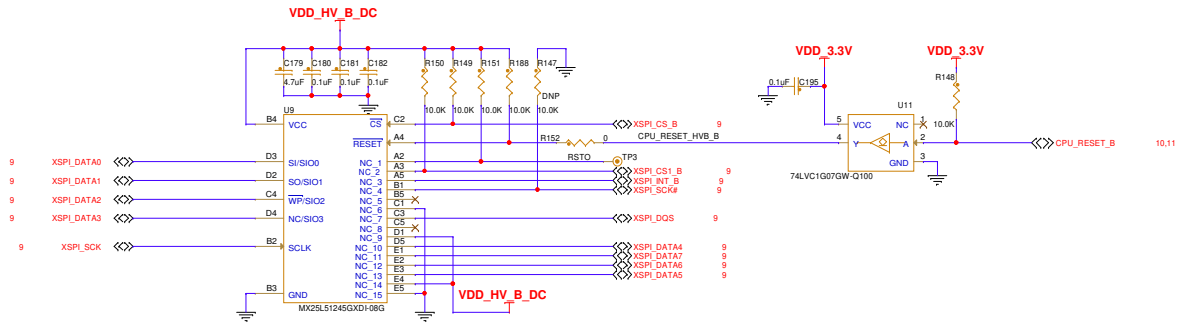
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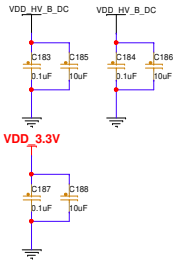
Page Title: 009 XSPI_MUXES_&_MEMORY

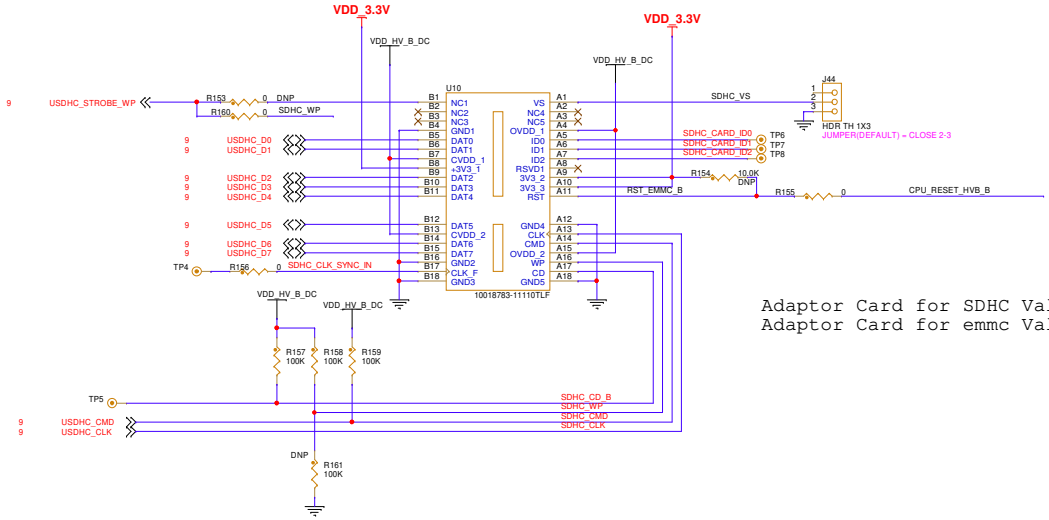
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XSPI Memory and SDHC Connector







Adaptor Card for SDHC Validation: 170-47572
Adaptor Card for emmc Validation: 170-31421

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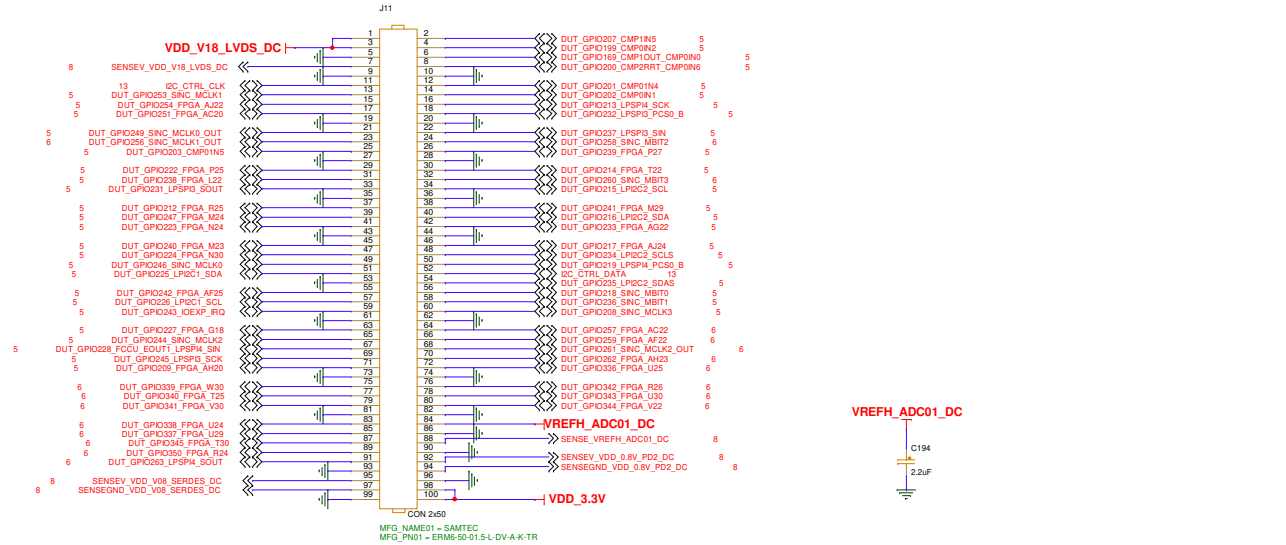
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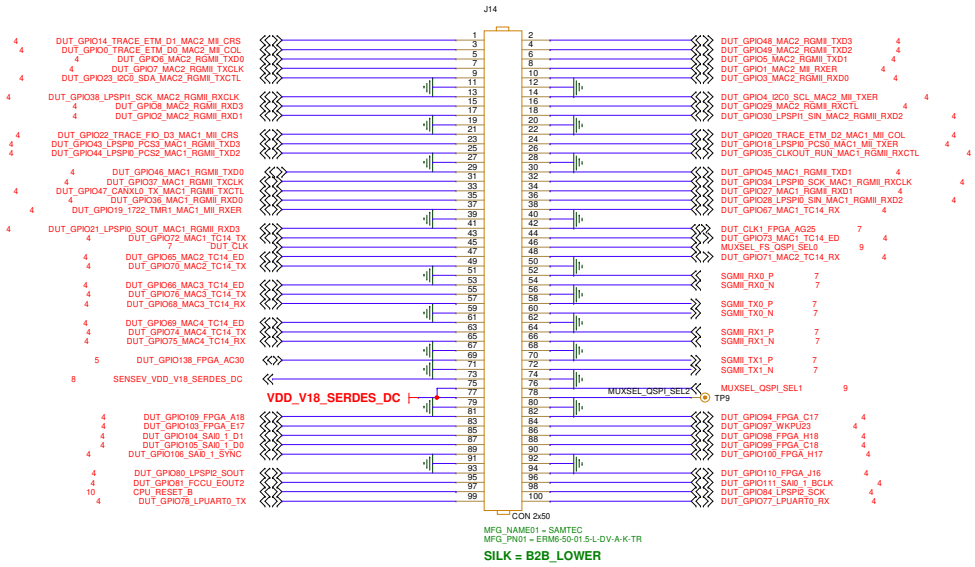
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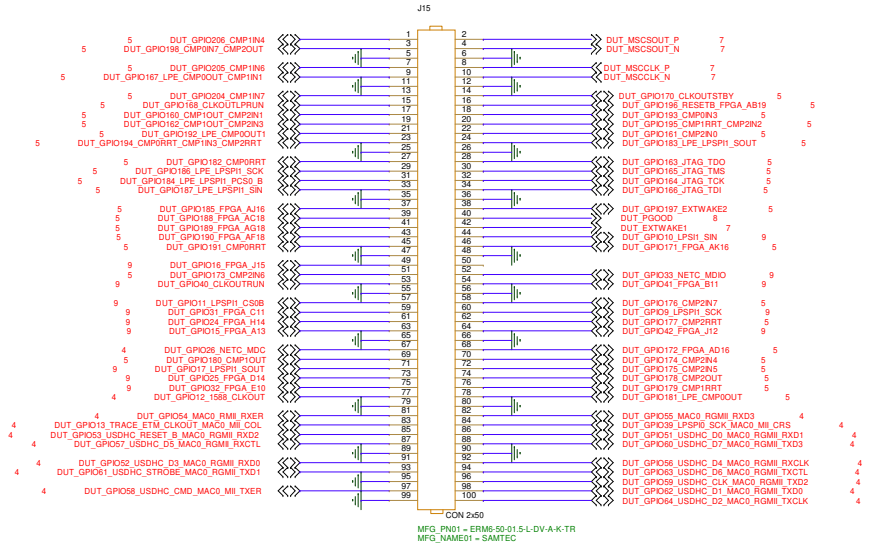
Board to Board Connector- Daughter Board





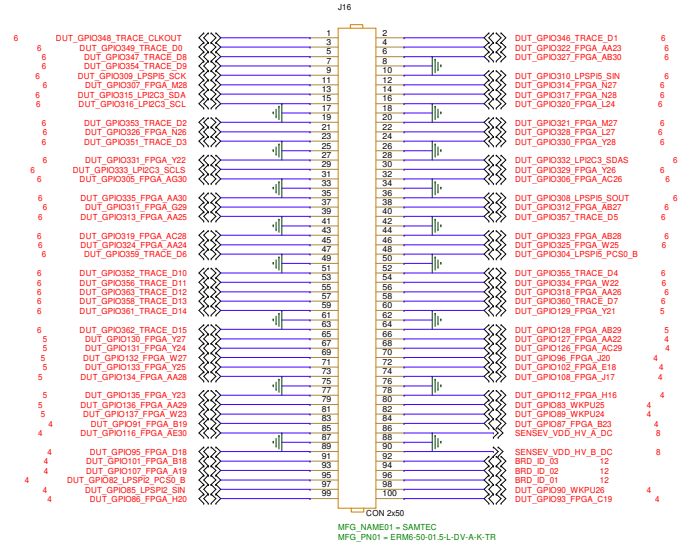
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Page Title:	011 S32K5XX_B2BConn_1	
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Board to Board Connector- Daughter Board



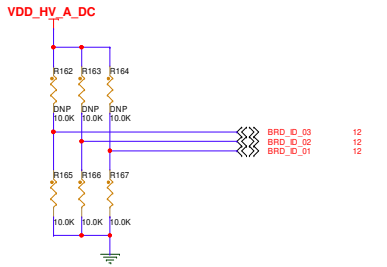
MFG_PN01 = ERM6-50-01.5-L-DV-A-K-TR
MFG_NAME01 = SAMTEC

SILK = B2B_LEFT



MFG_NAME01 = SAMTEC
MFG_PN01 = ERM6-50-01.5-L-DV-A-K-TR

SILK = B2B_RIGHT



Board ID[3:1]	Variant
000	3.3V XSPI FVAL DC <<
001	
010	
011	
100	1.8V XSPI FVAL DC
101	
110	
111	

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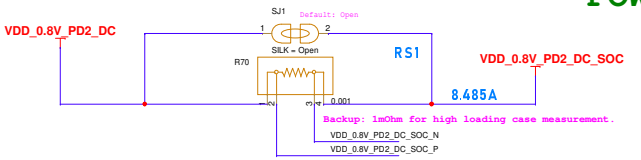
Drawing Title: X-S32K5XX-437-DC

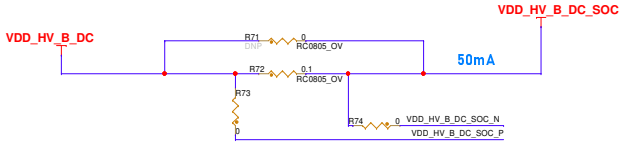
Page Title: 012 S32K5XX_B2BConn_2

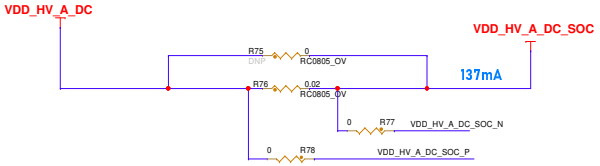
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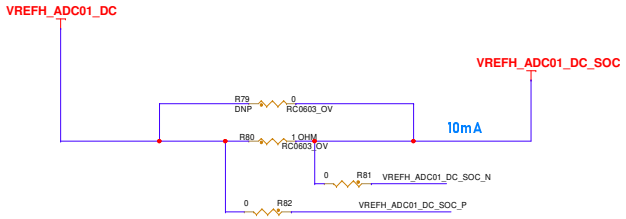
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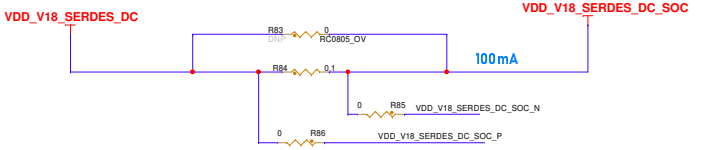
POWER MEASUREMENT CONNECTOR

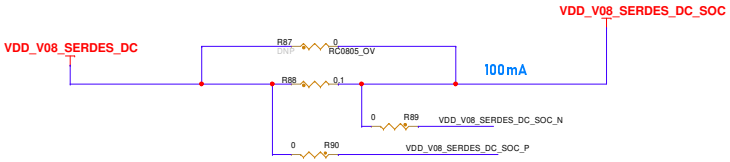


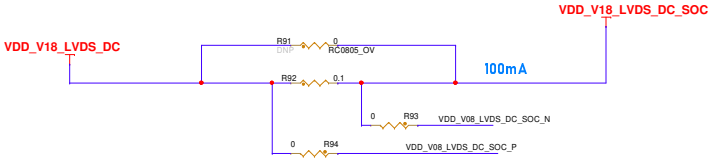


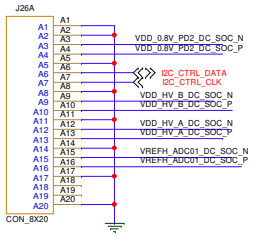


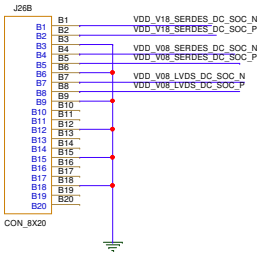


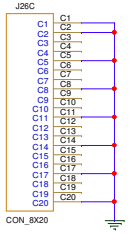


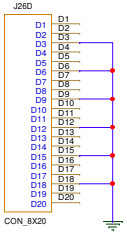


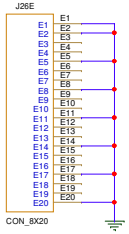


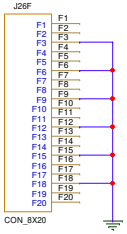


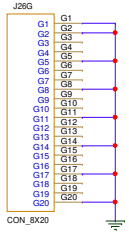


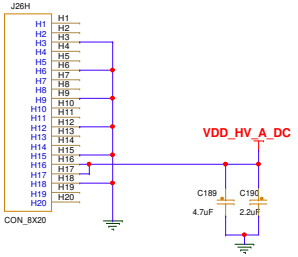












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